Week 2 Report: DINO from First Principles

1. What this was about

The goal this week was to understand how a computer can learn to recognise useful things in pictures without anyone telling it what is in them. Learning without labels like this is called self-supervised learning. Building a small version of an idea is one of the best ways to understand it, so a mini version of DINO was implemented and trained on 5,000 CIFAR-10 images. CIFAR-10 is a well-known collection of tiny 32x32 pictures of things like cats, cars, birds and airplanes. The model was a small vit, and training lasted 20 epochs.

2. The idea in plain words

DINO works like a game that a network plays against a copy of itself. One photo is turned into several differently cropped and coloured versions, and these are shown to two networks, called the student and the teacher. The student has to give the same answer as the teacher for that photo, even though the two saw different crops. The only dependable way to do this is to notice what is really in the picture, rather than accidental details such as exact colour or position. That is where the useful features come from, and no label is needed anywhere.

Student and teacher

Both networks have exactly the same design. The student learns the normal way, through backpropagation . The teacher is never trained directly. Instead, it follows the student. Each network ends in a small output section, called a head, which produces 2,048 numbers. A softmax then turns those numbers into probabilities, which are simply scores that add up to 1. The loss, which is the number that measures how wrong the student currently is, is the cross-entropy between the teacher's probabilities and the students'. In simple terms, it is small when the two agree and large when they disagree.

EMA (momentum) teacher

After every training step, the teacher moves a tiny bit closer to the student, using this rule: teacher = m * teacher + (1 - m) * student. EMA stands for exponential moving average. The value m starts at 0.99 and rises to about 0.9999 by the end. Since m is so close to 1, the teacher barely changes at each step, so it behaves like a slow, smoothed out version of the student. This means the targets the student aims at change gently rather than jumping around after every update.

Multi-crop

Every image is turned into six views: 2 large global crops (32x32) and 4 small local crops (16x16). All of them get random flips, colour changes, grayscale, blur and, sometimes, solarisation (a partial colour inversion). The teacher only sees the 2 global crops, while the student sees all 6. This forces the student to work out that a tiny piece of an image belongs to the same thing that the teacher sees in the whole picture.

Centering and sharpening

There is an easy way for the networks to cheat: both can output the same thing for every image, so the student and teacher always agree perfectly while learning nothing. This failure is called collapse, and DINO prevents it with two opposite nudges. Centering subtracts a running average of the teacher's outputs, which stops any single output from dominating and pushes the teacher towards a flat, even

spread of probabilities. Sharpening uses a low teacher temperature ,which makes the teacher decisive and peaky and pushes it towards picking one winner. Either one alone fails in its own way, but together they keep each other in check.

3. What was built

The notebook follows the whole pipeline step by step: getting the dataset, multi-crop augmentation, a small ViT used for both student and teacher, the DINO loss with centering and sharpening, the EMA update, training, and finally the embedding and attention visualisations. An embedding is just the list of numbers a network produces to describe an image. CIFAR-10 does come with labels, but the training data loader throws them away on purpose. They are only used afterwards, to colour the plots and to run a k-NN check. In a k-NN check, each test image is given the most common class among the training images whose embeddings are closest to its own. If the embeddings are good, similar images sit close together and the guesses are mostly right, so the accuracy acts as a score for the features.

The setup was as follows. The data was 5,000 CIFAR-10 training images (no labels used in training), plus 2,000 test images for the k-NN check. The model was a ViT with 4x4 patches, dimension 192, 6 blocks and 3 attention heads, which adds up to 2.69M parameters, plus a head with 0.69M parameters. The output size K was 2,048, whereas the paper uses 65,536. Each image produced 2 global crops (32x32) and 4 local crops (16x16). Training ran for 20 epochs with a batch size of 64, using the AdamW optimiser with a learning rate of 2e-4 and weight decay of 0.04. The teacher's EMA momentum went from 0.99 up to about 0.9999, and its temperature was fixed at 0.04. The student temperature was 0.1 and the centering momentum was 0.9.

4. What was observed

Sanity checks first

Before any training, a few checks made sure the plumbing was correct. The student produced 384 outputs per batch (64 images x 6 crops) and the teacher produced 128 (64 x 2), exactly as expected. At the start, the two networks matched perfectly (a difference of 0.0), and gradients reached the student but never the teacher. The EMA update was also checked by hand: the result matched the formula to within about 1e-8, m = 1 left the teacher untouched, and m = 0 turned it into an exact copy of the student.

The first run collapsed

This was the most instructive part. The first attempt used a learning rate of 5e-4 and a teacher temperature that rose from 0.04 to 0.07. The loss went up every epoch, which looked wrong, and the diagnostics explained why. Teacher entropy (a measure of how spread out the teacher's probabilities are) climbed from 6.6 to 7.54, out of a maximum of 7.62, which means a completely flat output. The student-teacher KL (a measure of how different the two sets of probabilities are) shrank to about 0.04, and k-NN accuracy stayed at roughly the untrained level. The teacher had stopped saying anything useful, so there was nothing for the student to learn. In other words, centering was winning over sharpening. The fix was to keep the teacher temperature at 0.04 throughout and to lower the learning rate to 2e-4. Everything from here on comes from that second run. This was a good reminder that the loss value alone says very little in DINO, and that entropy and KL are the real health indicators.

Training behaviour

In the working run, the loss rose at first, from 7.01 in epoch 1 to a peak of 7.29 in epoch 3, and then fell steadily to 6.66 by epoch 20. The early rise is not a bug. The loss is roughly the teacher entropy plus the student-teacher KL, and the teacher entropy was still going up at that stage (from 6.32 to 6.88). After that, entropy settled at around 5.9, well below the 7.62 maximum, while the KL slowly grew from 0.40 to about 0.78. This shows that the teacher kept making confident, varied predictions and the student always had something to learn from. No collapse happened.

The distance between the student's and teacher's weights grew to about 1.35 by epoch 6, and then eased down to about 1.07 as the momentum got close to 1. It never reached zero, which is exactly what is expected from a teacher that trails behind the student.

Put side by side, the main numbers look like this. The DINO loss went from 7.01 in epoch 1, to a peak of 7.29 in epoch 3, to 6.66 in epoch 20. Teacher entropy went from 6.32 in epoch 1, to a peak of 6.88 in epoch 3, to 5.88 in epoch 20, against a maximum possible of 7.62. The student-teacher KL began at 0.70 in epoch 1, dipped to 0.40 in epoch 3, and ended at 0.78 in epoch 20. The weight distance peaked at about 1.35 in epoch 6 and finished at 1.07. For k-NN accuracy, the untrained ViT scored 25.7%, and after training the teacher reached 28.9% and the student 30.6%.

Did it learn anything useful?

It did, but only a little, and it is important to be honest about how little. A random, untrained ViT already scores 25.7% on the k-NN test, because even random features keep colour and texture information. After training, the teacher reached 28.9% and the student 30.6%, which is a gain of about 3.2 and 4.9 points over the untrained network. With only 2,000 test images, a single point of accuracy is within noise, so the improvement is real but small. It also climbed steadily across the epochs rather than jumping, which fits the picture shown by the entropy curve.

One thing differs from the paper: here the student ended up slightly ahead of the teacher, while in the paper the teacher is usually better. A possible reason is that, in such a short run, the teacher (being a slow average) is still catching up with a student that is improving quickly. This was not tested, so it remains only a guess.

Embeddings (PCA and t-SNE)

PCA and t-SNE are two ways of squashing the 192-number embeddings down to a flat 2D picture so that they can be plotted. Honestly, these plots are not dramatic. The classes overlap heavily in both. There is a rough split between vehicles and animals in the trained model, but a similar loose split also appears in the untrained one, so not much should be concluded from eyeballing the pictures. The k-NN numbers are the better evidence here, and they say that the trained features are a bit better, not that they are cleanly separated.

Attention maps

Attention maps show where the model is looking. Here, they show where the CLS token looks in the last block. In the trained teacher, the attention is more concentrated: on the bird it lands on the head and body, on the cat on the face and body, and on the car on the roof. However, it is far from the clean object outlines shown in the paper. For the airplane it mostly sits on the ground, and head 3 often lights up the background. The untrained network is also surprisingly object-shaped, spreading its attention along edges and the outline of the animal. So a difference does exist, but at 32x32 pixels and 20 epochs it is subtle, and it cannot fairly be called a clear emergence of segmentation.

5. Ablation: what happens without the EMA teacher?

An ablation means removing one ingredient to see what it was really doing. Here, exactly one thing was changed, while the seed, the data and every other setting stayed the same. Instead of using the EMA, the teacher was simply set to a copy of the student after every step (m = 0).

The normal EMA run finished with a loss of 6.659 and a teacher entropy of 5.88, with k-NN accuracy of 28.9% for the teacher and 30.6% for the student. The no-EMA run finished with a loss of 7.643 and a teacher entropy of 7.62, which is the maximum, with 26.6% k-NN accuracy for both teacher and student. For reference, the untrained ViT scored 25.7% for both.

The difference in behaviour is much bigger than the difference in accuracy. Without the EMA, training was erratic. The loss and entropy swung around wildly: entropy was as low as 1.5 in epoch 3, then jumped to 5.9 in epoch 9, dropped to 2.2, and went up again. From epoch 13 onward the teacher entropy climbed to 7.62, which is exactly a uniform output. By epoch 17 the model had fully collapsed, with a KL near 0.02. The EMA run stayed stable the whole way through. This matches the paper, where a plain student-copy teacher fails.

What is surprising is that the no-EMA k-NN accuracy did not crash. It wandered between 22.5% and 28.4% and ended at 26.6%, only about one point above untrained. A likely explanation is that k-NN uses the 192-number backbone embedding, while the collapse happens in the 2,048-number head output, so a collapsed head does not completely wipe out the backbone's features. This has not been verified, so it should be treated as a probable reason rather than a proven one. It also means that k-NN on its own would have made the no-EMA run look fine, and it was the entropy curve that exposed the problem.

6. Limitations and takeaways

- Everything here is small: 5,000 images, a 2.7M-parameter model, 20 epochs, K = 2,048, and a single random seed. The results demonstrate the mechanism but are not a reproduction of the paper, and differences of a point or two are within noise.
- Tuning mattered. The first set of hyperparameters (the settings chosen before training) collapsed, and the learning rate and teacher temperature were changed after that, so the final settings are not untouched defaults.

The biggest takeaway is how much of DINO is about not collapsing. The idea of matching a teacher is simple, but making it work depends on the EMA teacher, centering and sharpening all staying in balance. The loss curve alone would have been misleading twice, once in the collapsed first run and once in the no-EMA run, and it was the entropy and KL diagnostics that showed what was really happening. Even at this tiny scale, self-distillation without labels produced features that were measurably better than random ones, which is the whole point of the exercise.